

Temporal significant wave height estimation from wind speed by perceptron Kalman filtering

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Abstract

Significant wave height estimates are necessary for many applications in coastal and offshore engineering and therefore various estimation models are proposed in the literature for this purpose. Unfortunately, most of these models provide simultaneous wave height estimations from wind speed measurements. However, in practical studies, the prediction of significant wave height is necessary from previous time interval measurements. This paper presents a dynamic significant wave height prediction procedure based on the perceptron Kalman filtering concepts. Past measurements of significant wave height and wind speed variables are used for training the adaptive model and it is then employed to predict the significant wave height amounts for future time intervals from the wind speed measurements only. The verification of the proposed model is achieved through the dynamic significant wave height and wind speed time series plots, observed versus predicted values scatter diagram and the classical linear significant wave height models. The application of the proposed model is presented for a station in USA.

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1. Introduction

Significant wave height predictions are preliminary prerequisites in any offshore engineering, planning, design, operation and maintenance studies. [Liu et al. \(2002\)](#)

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illustrate that whether typical wave prediction models based on the concept of a wave energy spectrum may have reached its limit or not. Although there are wave height estimation models, they do not take into account the meteorological and atmospheric variabilities entirely and consequently their results can be considered only as a general approximate trend to the overall dynamic behaviour (Tucker and Pitt, 2001; Komen et al., 1994; Schwab et al., 1984; Liu et al., 1984; Resio, 1981; Hughes and Jensen, 1986). In practical works, it is not possible to consider all the astronomical, atmospherical and meteorological variables for detailed estimations or predictions. In offshore engineering studies only discrete time measurements of significant wave height and many meteorological variables are available. It is, therefore, necessary to try and develop models either for the so-called significant wave height estimations from simultaneous readily available wind speed or time-wise prediction models that provide future significant wave height amounts at a site. Although there are time series modelling of significant wave height records, they depend on the overall statistical parameters of the records and they do not update themselves in order to accord with the new situations or measurements (Rodriguez et al., 2002; Forristall et al., 1996).

A simple three-layered feed forward type of neural network is developed by Deo et al. (2001) to obtain the output of the significant wave heights and average wave periods from the input of generating wind speeds. They managed to obtain model results with much less relative errors by using large sampling intervals, e.g. 1 week. However, this sampling interval is too large for real time predictions. The better real time prediction is proposed by this study for hourly, three hourly and seven hourly values of significant wave height measured simultaneously with the wind speed. The model results obtained in this study is with the 10% relative error.

Extensive review of the significant wave height modelling indicates that in practical studies, most often, a linear model between the significant wave height as dependent and one or more meteorological factors as independent variables are considered and least squares analysis leading to classical regression technique are employed. There are many restrictive assumptions in all the regression approaches and therefore more than physical restrictions procedural ones play role in the final estimation or prediction results. For instance, the requirement of Gaussian distribution of the residuals in any regression approach render such an approach into a rather difficult treatment in verification of such a distribution. To the best of author's knowledge, only in very rare studies the researchers look for the verification of the basic assumptions. More likely, they overlook the assumptions or ignore their significance in modelling as if the data are suitable for the assumptions set.

It is the main purpose of this paper to avoid these drawbacks and make dynamical adaptive significant wave height predictions for future time intervals (hours, days, months, years). For this purpose, a linear system equation is considered together with a measurement equation and their combined processing with the incoming of new measurements provide the future predictions in an objective manner.

2. Regression versus Kalman filter technique

Regression approach has been used invariably in many ocean wave studies that seek relationship between the significant wave height and wind speed. It is preferable to predict significant wave height from given wind speed values. The regression method (RM) brings along some restrictive assumptions, which are not strictly considered, in many applications. The basis of the method is curve-fitting procedure to scatter diagram between two or more variables. Unfortunately, in RM the scatter of the points on the Cartesian coordinate system is not cared for, but rather the least squares procedure is applied so as to minimize the sum of square errors. Of course, it is necessary to have a pre-determined mathematical relationship, which needs parameter estimations only. This implies that the dynamic of the system is already restricted by a specific determination expression. The following points can be cited among the drawbacks in any RM application for significant wave height prediction.

- (1) The scatter diagram does not take into consideration the time evolution of the phenomenon in the prediction sequence, but provides global estimations in the sense of a deterministic relationship between two variables (significant wave height and wind speed).
- (2) The deviations of the scatter of points from the fitted curve will have zero average, and the variance of these deviations is assumed as constant. However, in actual scatter diagrams most often the variance is not constant and changes depending on the independent variable value. For instance, there is no clear cut pattern of relationship between the significant wave height and wind speed as obvious from the scatter diagram in Fig. 1, which is based on measured data from 46002 station at Oregon, USA.

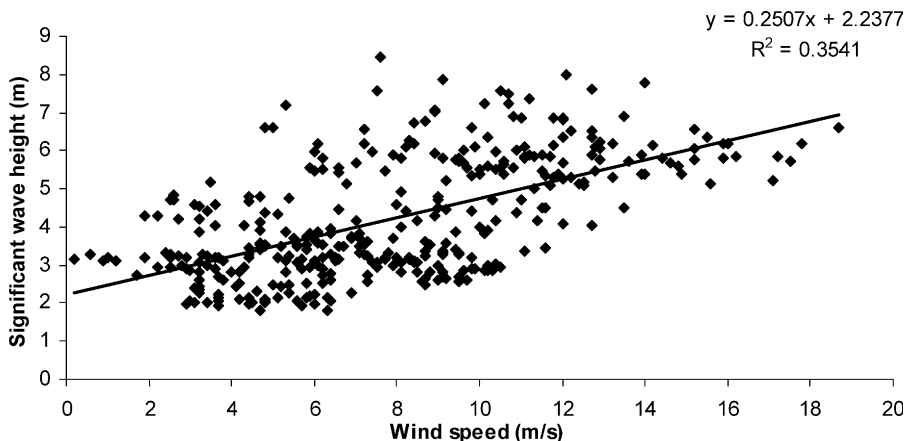


Fig. 1. Scatter diagram of significant wave height and wind speed.

- (3) In such a very scattered diagram the fitting of a regression curve may pass close to a certain percentage of points, but this cannot be accounted for the validity of the method, if the basic assumption requirements are not satisfied. Additional discussion is given in the Section Application of this paper.
- (4) The prediction errors are expected to abide with a Gaussian distribution function which is not the case in many practical studies.
- (5) The prediction errors are also expected to be independent from each other, i.e. completely random (noise).

Recognizing the above limitations, the present study attempts to address significant wave height prediction problem from a new perspective, by coupling artificial neural networks for estimating the connection parameters and the Kalman filter for predictions. Therefore, the study is expected to form an important contribution to wave studies, and the outcomes could provide further evidence to the potential of the modern artificial neural networks and the Kalman filtering techniques in coastal and offshore engineering applications.

Kalman filtering (KF) is an adaptive modelling of state variables some of which are measured, and hence, the overall state variables can be estimated sequentially. The fundamentals of the Kalman filtering technique are explained by Kalman (1960) and Gelb (1974). In the following, a brief description of KF is presented with some explanations. It relates current state variable, $\mathbf{X}_k$, to previous time state variable $\mathbf{X}_{k-1}$ through a transition matrix $\Phi_{k/k-1}$ which converts the state variables from time, $k-1$, to the next, k , in addition to system error vector $\mathbf{W}_{k-1}$ as

$$\mathbf{X}_k = \Phi_{k/k-1}\mathbf{X}_{k-1} + \mathbf{W}_{k-1} \quad (1)$$

This is the system equation, and the dimensions of state and error vectors are $(n \times 1)$ where n is the number of state variables.

Another ingredient in a KF technique is a measurement equation, which transforms the state vector $\mathbf{X}_k$ into a measurement vector, $\mathbf{Z}_k$ as

$$\mathbf{Z}_k = \mathbf{H}_k\mathbf{X}_k + \mathbf{V}_k \quad (2)$$

where $\mathbf{H}_k$ is an $(m \times n)$ matrix representing the connection between the measurement and the state vectors at time k , and $\mathbf{V}_k$ is measurement errors vector of dimension $(m \times 1)$. The KF model works such that the system and measurement errors are independent structurally and mutually with zero means and finite variances, $\mathbf{Q}_k$ and $\mathbf{R}_k$, respectively. Predictions are obtained by successive use of the system and measurement equations. First, the system equation gives prediction $\hat{\mathbf{X}}_{k/k-1}$ at time k from available information at time $k-1$, and then this prediction is substituted into Eq. (2) for measurement error calculation as $\mathbf{Z}_k - \mathbf{H}_k\hat{\mathbf{X}}_{k/k-1}$. In order to find the final state prediction, this measurement error is combined with $\hat{\mathbf{X}}_{k/k-1}$ prediction through a Kalman gain vector, $\mathbf{K}_k$ as

$$\hat{\mathbf{X}}_{k/k} = \hat{\mathbf{X}}_{k/k-1} + \mathbf{K}_k(\mathbf{Z}_k - \mathbf{H}_k\hat{\mathbf{X}}_{k/k-1}) \quad (3)$$

Herein, the Kalman gain factor is obtained in such a way that the prediction mean square error is minimized which yields to (Kalman, 1960)

$$\mathbf{K}_k = \mathbf{P}_{k/k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k/k-1} \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \quad (4)$$

With all the aforementioned symbols, the execution steps of KF is presented in detail by different researchers as (Kalman, 1960; Gelb, 1974)

$$\hat{\mathbf{X}}_{k+1/k} = \Phi_{k+1/k} \hat{\mathbf{X}}_{k/k} \quad (5)$$

$$\mathbf{P}_{k+1/k} = \Phi_{k+1/k} \mathbf{P}_{k/k} \Phi_{k+1/k}^T + \mathbf{Q}_k \quad (6)$$

$$\hat{\mathbf{X}}_{k/k} = \hat{\mathbf{X}}_{k/k-1} + \mathbf{K}_k (\mathbf{H}_k \mathbf{X}_k + \mathbf{V}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k/k-1}) \quad (7)$$

and

$$\mathbf{P}_{k/k} = \mathbf{P}_{k/k-1} - \mathbf{K}_k \mathbf{H}_k \mathbf{P}_{k/k-1} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k/k-1} \quad (8)$$

The application of KF technique goes in sequence by successive execution of Eqs. (5) and (8) together with Eq. (4).

It is, therefore, suggested in this paper to try and develop a model relating wind speed (S) to significant wave height (H_s) on the basis of KF system equation. For significant wave height prediction in all the aforementioned KF equations, the state variable is considered as composed of two variables, namely, the significant wave height, H_s and the wind speed, S . Hence, the system equation of state variables similar to Eq. (1) can be written explicitly as,

$$\begin{bmatrix} S \\ H_s \end{bmatrix}_k = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} S \\ H_s \end{bmatrix}_{k-1} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}_{k-1} \quad (9)$$

Herein, a_{11} , a_{21} , a_{12} and a_{22} are the transition matrix elements, which show the contributions of the previous H and S values to the current amounts.

3. Neuron network perceptron

In Eq. (10), H_s and S refer to measured past values and the transition matrix elements must be calculated from these measurements. Hence, prior to the complete application of the KF technique, it is necessary to calculate a_{11} , a_{21} , a_{12} and a_{22} values from a given set of H_s and S measurements. For this purpose, the transition matrix mechanism can be considered explicitly as,

$$S_k = a_{11} S_{k-1} + a_{21} (H_s)_{k-1} + (w_1)_{k-1} \quad (10)$$

and

$$(H_s)_k = a_{12} S_{k-1} + a_{22} (H_s)_{k-1} + (w_2)_{k-1}. \quad (11)$$

It is possible to map these two equations into the neural network form as in Fig. 2 where in the input layer there are two neurals with variables $(H_s)_{k-1}$ and S_{k-1} . The output layer has also two neurals with output variables $(H_s)_k$ and S_k .

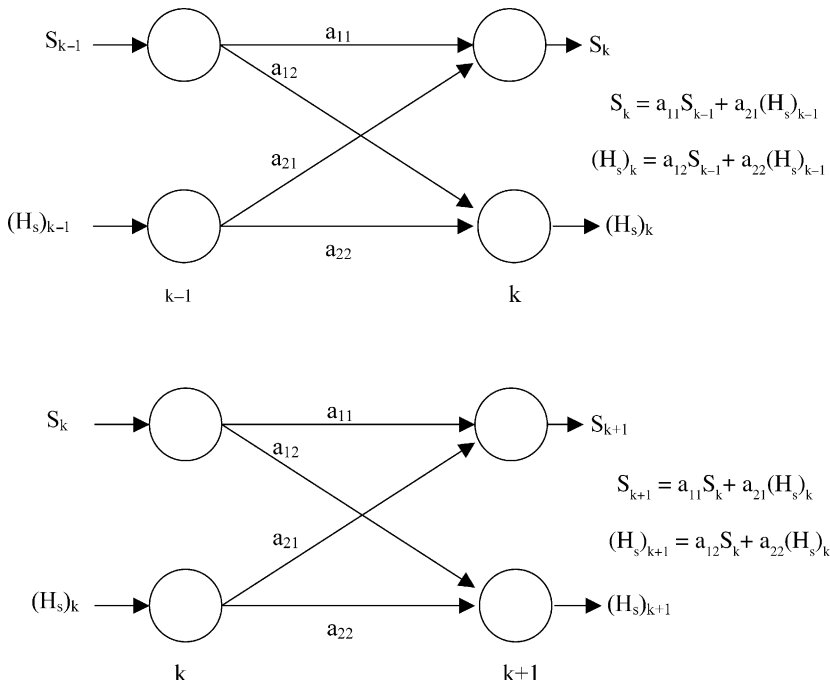


Fig. 2. Perceptron configuration.

These two layer neurals are also connected to each other through a net of transition matrix weights. Since, there is not a hidden layer, this system is referred to as the perceptron (Holland, 1975). Consequently such an architecture causes linear transformations between the input and output variables. For any given pairs of $((H_s)_{k-1}, S_{k-1})$ and $((H_s)_k, S_k)$, the weights are obtained by considering back propagation mechanism in the training. Once the best weights (a_{11} , a_{21} , a_{12} and a_{22}) are obtained for a given pairs of (H_s) and S measurements, then the next measurement pairs are considered, and with the repetition of the same procedure, a new set of a_{11} , a_{21} , a_{12} and a_{22} values are obtained. This is achieved until the end of pre-selected length (training period) of measurement sequences. Since (H_s) and S vary by time, so will the weight estimates, and consequently, each weight (a_{11} , a_{12} , a_{21} and a_{22}) will have its own sequence of change with time.

Once the transition matrix elements are determined then the regular KF procedure with measurement equation is applied according to the explanations in the two previous sections.

4. Application

Since, for the prediction purpose a combined use of perceptron and KF techniques are considered, this approach will be referred to as the perceptron KF (PKF)

method in this paper. The implementations of PKF method are performed for station 46002 which is located in the Coos Bay at Oregon, USA. This station is owned and maintained by National Data Buoy Center.

The perceptron part of the methodology is applied for 15 days of the month January, 2003 and finally sequences of coefficients (a_{11} , a_{21} , a_{12} , a_{22}) estimation are obtained. The question is now which weight must be chosen for the best prediction of H_s and S . Although the arithmetic averages are the first approximations in practical studies, herein, the most frequently occurring weight values, i.e. modes are adopted for prediction purpose (Table 1). Hence, the transition matrix elements in the PKF system model are taken as modes of the weights that are calculated by perceptron model during the training period. Such a selection will not result in serious problems, since the mode value frequency of appearance of the coefficient within the measurement length is necessarily the best indicator. In order for the mode coefficient to occur repeatedly, data set is sufficient. The classical RM techniques possess an important limitation by requiring continuous long-term data sets. In this context, the use of artificial neural networks for coefficient determination with the mode is effective than any other technique. The whole data set is thought of two parts, namely, 15 days of training period through perceptron methodology and then 16 days of prediction part through the PKF procedure. Training part yields the elements of system matrix. For the study site, the system matrix with mode values of parameters are shown in Table 1.

These parameters correspond to the transition matrix in system Eq. (1). In the PKF application part, first of all the system and measurement error variances are adopted arbitrarily as,

$$Q_0 = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \text{ and } R_0 = [1] \quad (12)$$

Herein, diagonal values are taken as high values in the expectation that through the PKF operation, the trace of these error matrices will decrease by time and stabilize after some time. The system prediction of significant wave height is achieved by the application of Eq. (5) where the initial state variable values are adopted as the last observed values of the training period.

Since, the main purpose of this study is to estimate significant wave height from wind speed measurements, the measurement matrix in Eq. (2) is considered as,

$$H = [1 \quad 0]. \quad (13)$$

Table 1
Calculated weights and predictions for station 46002

Station 46002	a_{11}	a_{21}	a_{12}	a_{22}	Average relative error (%)
Hourly	1.02	−0.02	0.02	1	7.46
3-h moving average	0.36	0	0.05	0.94	4.72
7-h moving average	1.90	−0.80	−0.07	1.00	3.59

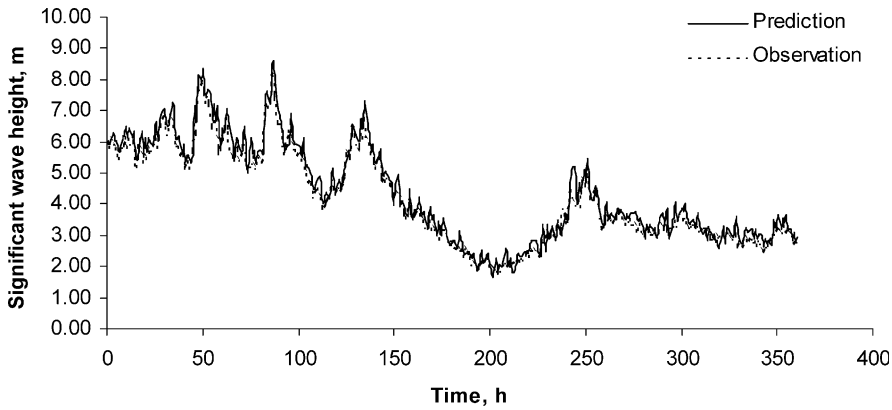


Fig. 3. Predicted and measured time series for hourly data.

At this stage, all the matrix elements are known on the right hand side of Kalman gain Eq. (4) and hence, it is possible to calculate K_k . Its substitution into Eq. (7) yields the updated estimate of state variable. Furthermore, the updated covariance matrix elements can be obtained from Eq. (8). Hence, the first round in the PKF application is completed, and its repetition with the coming new measurements gives updated subsequent state variable estimations.

The numerical application of this procedure in the study area gave the significant wave height predictions for hourly, 3-h moving average and 7-h moving average data set is shown in Figs. 3, 5 and 6, respectively. Model verification for hourly data is also shown in Fig. 4. It is obvious that there is a very good agreement between observed and PKF predicted values.

The comparison of RM and PKF models can be achieved comparatively on the basis of error definitions and graphical representations. Three numerical error com-

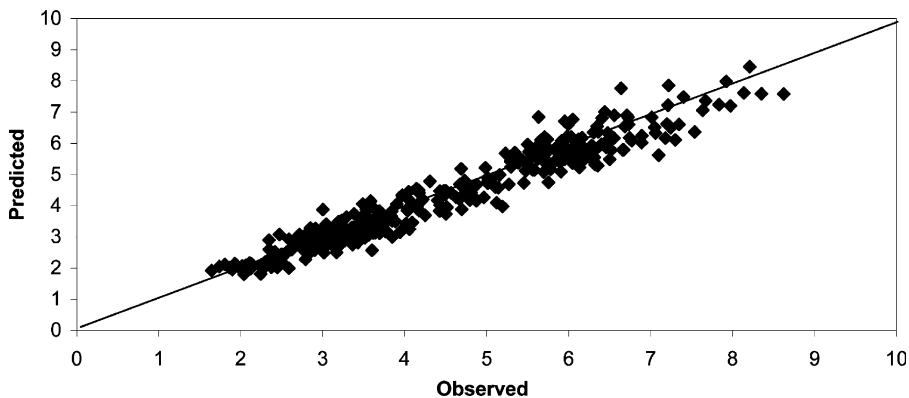


Fig. 4. Model verification for hourly data.

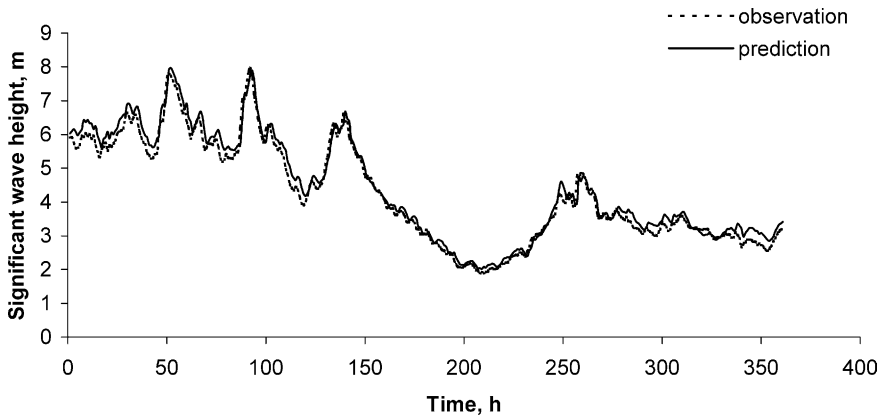


Fig. 5. Predicted and measured time series for 3-h moving average data.

parison is employed in this paper, namely, average relative error percentage which is one of the numerical error comparison is employed in this paper and defined as follows,

$$(\text{AREP})_i = 100 \frac{1}{n_p} \sum_{i=1}^{n_p} \frac{|(H_s)_{pi} - (H_s)_{mi}|}{(H_s)_{mi}} \quad (14)$$

where $(H_s)_{pi}$ and $(H_s)_{mi}$ are the predicted and measured concentration values at hour i . The error term for each day can be calculated as the terms in the summation sign of all these error formulations. The comparison of each method for last 2 days is presented in Fig. 7. As can be seen from this figure, in majority of predictions PKF relative error is less than RM prediction.

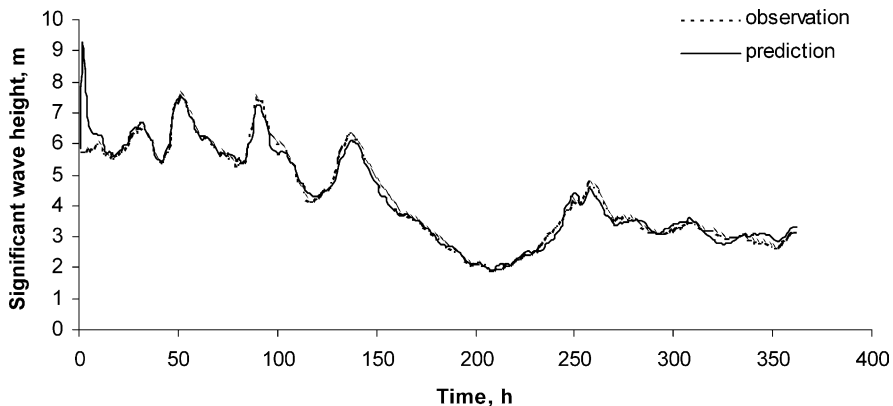


Fig. 6. Predicted and measured time series for 7-h moving average data.

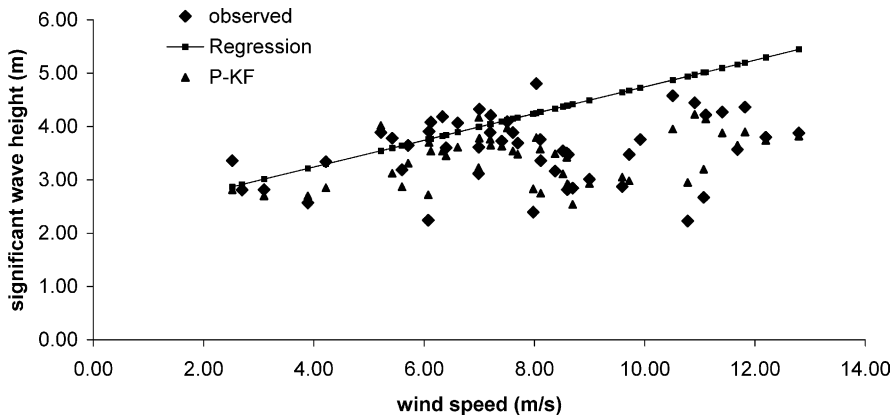


Fig. 7. Scatter diagram comparison.

5. Conclusions

Significant wave height predictions are preliminary requisites for proper future planning, operation and maintenance of offshore structures. There are many prediction procedures in practice, but they are based on the long-term parameter average estimations by classical RM as empirical or rational formulations. The best methodology is to identify and construct models based directly on the significant wave height and wind speed measurements. In this paper, a dynamic adaptive model is assumed to be valid between two successive time instances of significant wave height and wind speed variables. Such a model with uncertainty components suits best the classical KF procedure for prediction. However, the elements of the transition matrix are estimated from the measurement sequences by the use of perceptron concepts from the neural networks methodology. Perceptrons are the simplest forms of artificial neural networks without any hidden layer. The transition matrix elements are estimated through such a perceptron method and then the significant wave height predictions are achieved by the Kalman procedure, and therefore, the overall method is referred to as the perceptron KF method in this paper.

The application of the PKF method is performed for significant wave height and wind speed measurements from USA in the Coos Bay at Oregon. Numerical error criteria and graphical representations have shown that the PKF approach is better than the classical RM procedures. Especially, PKF works on a dynamic basis with future predictions from current measurements of the state variables. Hence, it does not depend on deterministic formulations curve, and consequently yields prediction scatter diagrams similar to observation scatter diagrams.

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